CS341 · COMPUTER GRAPHICS

# Calculations Solutions

Practice on every calculation type

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## 01 Solutions Q1–Q3

### Q1

a) $AR = \dfrac{\frac{768}{9}}{\frac{1024}{12}} = \dfrac{85.33}{85.33} = 1$

b) $AR = \dfrac{\frac{600}{8}}{\frac{800}{10}} = \dfrac{75}{80} = 0.9375$

### Q2

x: $\dfrac{1280}{32} = 40$ pixels per cm

y: $\dfrac{960}{24} = 40$ pixels per cm

$AR = \dfrac{40}{40} = 1$

### Q3

pixels = 1024 × 768 = 786,432

a) 786,432 bits ÷ 8 = 98,304 bytes

b) 786,432 × 8 bits ÷ 8 = 786,432 bytes  (768 KB)

c) 786,432 × 24 bits ÷ 8 = 2,359,296 bytes  (2.25 MB)

## 02 Solution Q4: vectors

a) $v = Q - P = (7 - 2,\ 1 - 5) = (5, -4)$  $|v| = \sqrt{(25 + 16)} = \sqrt{41} \approx 6.4$

b) $|a| = \sqrt{(1 + 4 + 4)} = 3$  $\hat{a} = (\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$

c) $a \cdot b = 1(2) + 2(-1) + 2(0) = 0$  → perpendicular

d) $a \times b = (a_y \cdot b_z - a_z \cdot b_y,\ a_z \cdot b_x - a_x \cdot b_z,\ a_x \cdot b_y - a_y \cdot b_x)$

$= (2 \cdot 0 - 2 \cdot (-1),\ 2 \cdot 2 - 1 \cdot 0,\ 1 \cdot (-1) - 2 \cdot 2)$

$= (2, 4, -5)$

check: $(2, 4, -5) \cdot a = 2 + 8 - 10 = 0$ ✓

e) $u \cdot w = -1$  $|u| = 1$  $|w| = \sqrt{2}$

$\cos\theta = -\frac{1}{\sqrt{2}}$  →  $\theta = 135°$

f) $c_\perp = (-c_y, c_x) = (2, 3)$

## 03 Solution Q5: points and lines

| Combination | Sum of coefficients | Valid point? |
|---|---|---|
| 0.5P + 0.5R | 1 | **Yes (midpoint)** |
| 2P − R | 1 | **Yes** |
| P + R | 2 | **No** |
| 0.2P + 0.3R | 0.5 | **No** |

```
b) P = (1, 2) + 0.75(8, 4) = (1 + 6, 2 + 3) = (7, 5)
```

```
c) b = B − C = (3, 4)          L(t) = (2, 1) + t(3, 4)
```

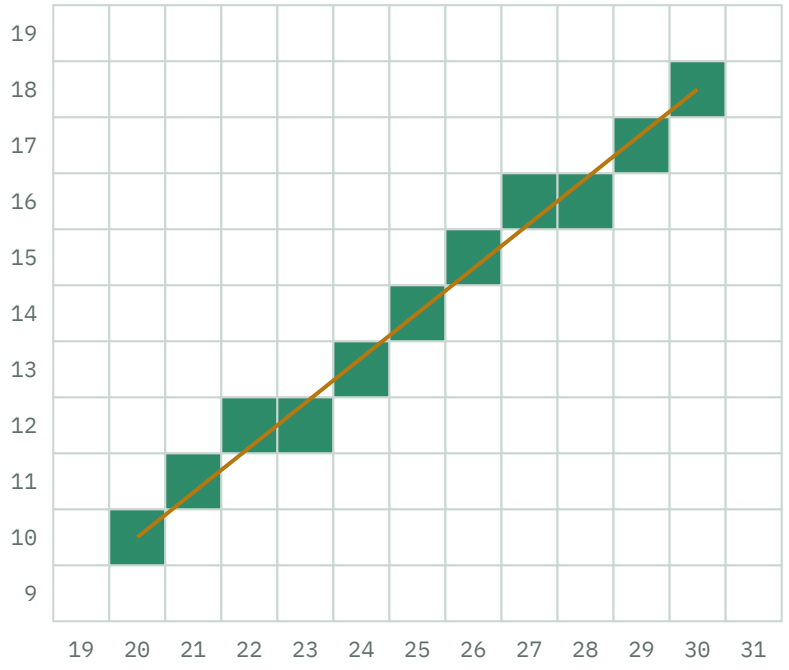
```
   n = b⊥ = (−4, 3)            −4(x − 2) + 3(y − 1) = 0
```

## 04 Solutions Q6–Q8: DDA

### Q6: (20,10) → (30,18)

```
dx = 10, dy = 8, steps = 10, xInc = 1, yInc = 0.8
```

| k | x | y | round(x) | round(y) |
|---|---|---|---|---|
| 0 | 20 | 10 | 20 | 10 |
| 1 | 21 | 10.8 | 21 | 11 |
| 2 | 22 | 11.6 | 22 | 12 |
| 3 | 23 | 12.4 | 23 | 12 |
| 4 | 24 | 13.2 | 24 | 13 |
| 5 | 25 | 14 | 25 | 14 |
| 6 | 26 | 14.8 | 26 | 15 |
| 7 | 27 | 15.6 | 27 | 16 |
| 8 | 28 | 16.4 | 28 | 16 |
| 9 | 29 | 17.2 | 29 | 17 |
| 10 | 30 | 18 | 30 | 18 |

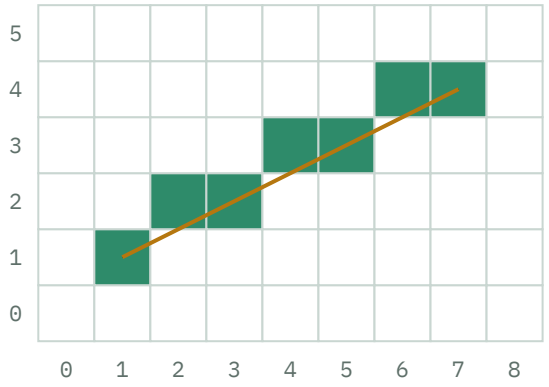


## Q7: (1,1) → (7,4)

$$dx = 6,\ dy = 3,\ steps = 6,\ xInc = 1,\ yInc = \frac{3}{6} = 0.5$$

$$round(a) = int(a + 0.5) \quad \rightarrow \quad 1.5 \rightarrow 2, \quad 2.5 \rightarrow 3, \quad 3.5 \rightarrow 4$$

| k | x | y | round(x) | round(y) |
|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 1 |
| 1 | 2 | 1.5 | 2 | 2 |
| 2 | 3 | 2 | 3 | 2 |
| 3 | 4 | 2.5 | 4 | 3 |
| 4 | 5 | 3 | 5 | 3 |
| 5 | 6 | 3.5 | 6 | 4 |
| 6 | 7 | 4 | 7 | 4 |




## Q8: (0,0) → (3,6)

$$dx = 3,\ dy = 6,\ m = 2 > 1,\ steps = 6,\ xInc = \frac{3}{6} = 0.5,\ yInc = 1$$

| k | x | y | round(x) | round(y) |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0.5 | 1 | 1 | 1 |
| 2 | 1 | 2 | 1 | 2 |
| 3 | 1.5 | 3 | 2 | 3 |
| 4 | 2 | 4 | 2 | 4 |
| 5 | 2.5 | 5 | 3 | 5 |
| 6 | 3 | 6 | 3 | 6 |

